

Learning objectives

In this lesson you will learn to:

- Explain the need for program translators
- Define what is meant by the terms ‘compiler’ and ‘interpreter’
- Compare the way in which interpreters and compilers translate source code into machine code
- Describe the advantages/disadvantages of each approach
- Select and justify which method of translation to use for a given purpose.

For more information and additional student activities see Topic 3.3 of the student book.

Explain the need for source code translators

- What do you think a source code translator does?
 - Converts the text in source, fit for a human, to machine code, fit for the computer
- Why do we need to do that?
 - Computers can't execute (run) the text that a human writes; they only execute binary.
 - The text is 'translated' into machine code.

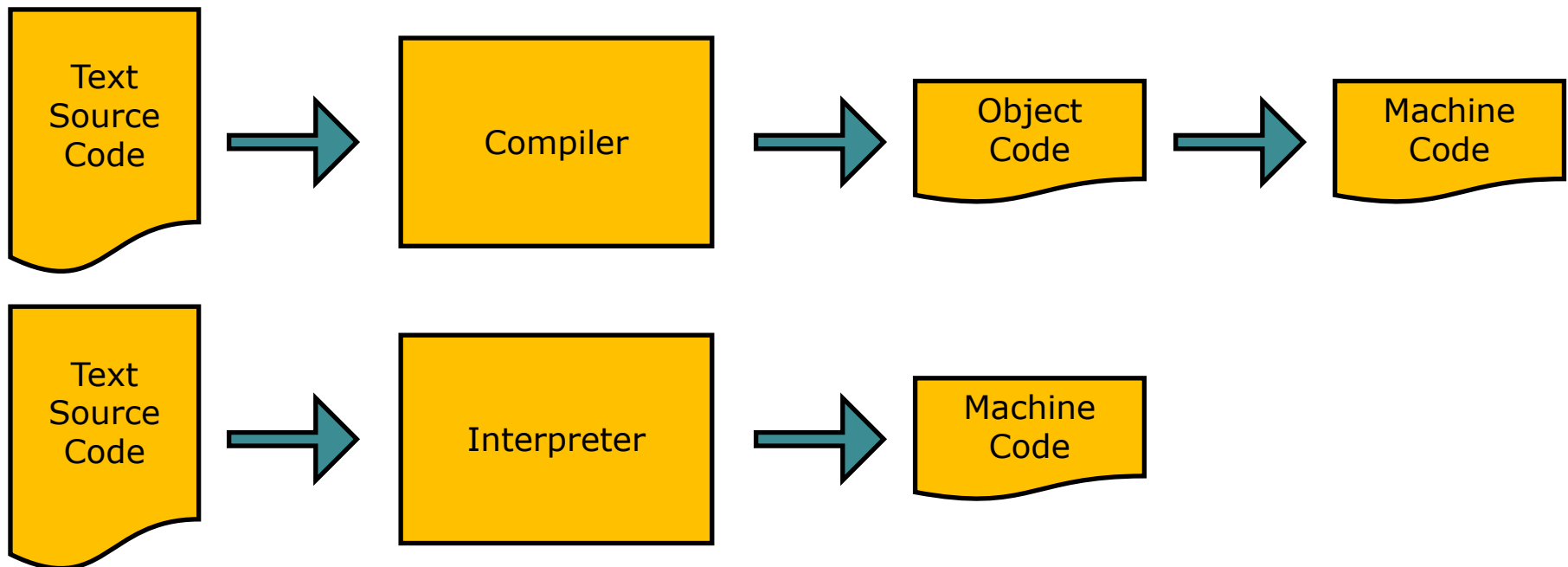
Key terms

The type of translator required depends on whether the source code is low-level or high-level.

- Machine code does not require a translator.
- Assembly code is low-level; it requires a translator named **assembler**.
- High-level languages require a translator named either **compiler** or **interpreter**.

How translators work

- Each translator works in a slightly different way.
- All of them take a source code file as input.
- All of them produce machine code.



Advantages and disadvantages of each approach

- Both compilers and interpreters have advantages and disadvantages.
- Considerations include:
 - Speed
 - Portability
 - Interactivity
 - Error-handling
 - Testing and debugging
 - Secrecy

How do you decide which to use?

- Whether a language is compiled or interpreted is not a reason to choose it.
- Normally, the choice of language is based on what you need to do.
- Sort out the objective first, then choose the best language to use.
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- The option of compiler or interpreter comes from that decision.

Wrap up: you have learned how to...

- ✓ Explain the need for program translators
 - To convert human-readable code to machine code
- ✓ Define what is meant by the terms 'compiler' and 'interpreter'
 - Compiler and interpreter translate high-level code
- ✓ Compare the way in which interpreters and compilers translate source code into machine code
 - Interpreters translate and execute one line at a time
 - Compilers translate a whole source file at once, to produce an executable.

and you have learned how to...

- ✓ Describe the advantages/disadvantages of each approach
 - Compiled executes quicker and is more secure
 - Interpreted is quicker to develop in and is more portable
- ✓ Select and justify which method of translation to use for a given purpose
 - Depends entirely on the objective of the task.